

1 0. Prof's Blueprint (High Priority)

- **Regression:** fit + predict; linearization + de-linearization; pick model from graph; Hessian + definiteness. Multivariable: $X^T X \bar{a} = X^T \bar{y}$, pseudoinverse. ML: features, decision boundary, σ -function.
- **Integration:** draw rectangles (α -formula), trapezoids, parabolas; compute rect/trap on uniform + non-uniform grids; Simpson on uniform. Accuracy estimates in h .
- **ODEs:** first-order IVP; exact-solution checks; Euler + midpoint steps + errors; **RK4 scheme** written down ($k_1..k_4$), no numbers needed. Accuracy estimates (no derivation).
- **Starred (may be excluded):** matrix/det basics, Laplace, Gaussian elimination, LU, det for $n > 3$, writing RK scheme.
- MCQ + written, 30 pts. 1 A4 sheet. Calculators OK; no phones.

2 1. Regression / Data Fitting (Th.1-6)

Regression $\neq$ interpolation: interpolation trusts every measurement and passes through ALL points; regression assumes noise and finds the best OVER-ALL trend, passing through (almost) none. Deviations $d_i = y_i - \phi(x_i)$, data (x_i, y_i) , $i = 0..n$ ($n+1$ points). **Error choices:** naive $S = \sum d_i$ (rejected: +, - cancel); abs $S = \sum |d_i|$ (rejected: kink at 0, no derivative); **squared** $S = \sum d_i^2$ (chosen: smooth everywhere, so "set gradient = 0" gives solvable linear equations; big misses punished quadratically). Example (Lec14): pts (1,2), (2,1), line through (1,1), (2,2): $d_0 = 1, d_1 = -1$: naive $S = 0$ (cancels!), abs $S = 2$, squared $S = 2$.

Linear regression

$\hat{y} = a_0 + a_1 x$, $S = \sum (y_i - a_0 - a_1 x_i)^2$. General fit $\phi(x, a_0..a_\ell)$: $\frac{\partial S}{\partial a_k} = -2 \sum_{i=0}^n (y_i - \phi(\dots)) \frac{\partial \phi}{\partial a_k} = 0$, $k = 0..\ell$ ($\ell+1$ equations). For the line this gives the two normal equations below. $\partial_{a_0}: \sum y_i - (n+1)a_0 - a_1 \sum x_i = 0$; $\partial_{a_1}: \sum x_i y_i - a_0 \sum x_i - a_1 \sum x_i^2 = 0$.

Divide by $n+1 \Rightarrow a_0 + a_1 \bar{x} = \bar{y}$, $a_0 \bar{x} + a_1 \bar{x}^2 = \overline{xy}$; Cramer gives

$$a_1 = \frac{\overline{xy} - \bar{x}\bar{y}}{\bar{x}^2 - \bar{x} \cdot \bar{x}}, \quad a_0 = \bar{y} - a_1 \bar{x}$$

with $\bar{x} = \frac{1}{n+1} \sum x_i$, $\bar{x}^2 = \frac{1}{n+1} \sum x_i^2$, $\overline{xy} = \frac{1}{n+1} \sum x_i y_i$.

Trap: $\bar{x}^2 \neq \overline{x^2}$; denominator = 0 iff all x_i equal (no fit possible).

Hessian + minimum test

$$H = \begin{pmatrix} S_{a_0 a_0} & S_{a_0 a_1} \\ S_{a_1 a_0} & S_{a_1 a_1} \end{pmatrix}, \quad S_{a_0 a_0} = 2(n+1), \quad S_{a_0 a_1} = 2 \sum x_i, \quad S_{a_1 a_1} = 2 \sum x_i^2$$

$$\det H = 4 \left[(n+1) \sum x_i^2 - \left(\sum x_i \right)^2 \right]; \min \iff S_{a_0 a_0} > 0 \text{ AND } \det H > 0$$

Intuition: H = curvature of the error bowl; positive-definite = bowl opens upward = exactly one bottom. **Cauchy-Schwarz** $\Rightarrow \det H \geq 0$; > 0 if ≥ 2 distinct $x_i \Rightarrow$ unique global min (strictly convex).

Multivariable + pseudoinverse

$\hat{y} = a_0 + a_1 x_1 + \dots + a_m x_m$, X is $(n+1) \times (m+1)$ with first col of 1's. $S = \|\bar{y} - X\bar{a}\|^2$.

$$X^T X \bar{a} = X^T \bar{y}, \quad \bar{a} = \underbrace{(X^T X)^{-1} X^T \bar{y}}_{X^+}$$

$X^+ =$ Moore-Penrose pseudoinverse ($= X^{-1}$ when X square). Idea: $X^T X \bar{a} = X^T \bar{y}$ projects $\bar{y}$ onto the column space of X (closest achievable fit); X^+ is that projection written as a formula. **Trap:** $X^+ X = I$ holds; $XX^+ = I$ generally does NOT.

ML: classification

Features = measured inputs (x_1, x_2) ; model outputs a score; plane $a_0 + a_1 x_1 + a_2 x_2 = 0 \cap x_1 x_2$ -plane = **decision boundary** (class 1 vs 2). Logistic $\sigma(z) = \frac{1}{1+e^{-z}}: \mathbb{R} \rightarrow [0, 1]$ squashes the score to a probability; classifier $\sigma(a_0 + a_1 x_1 + a_2 x_2)$.

3 2. Linearization

Goal: transform $(\bar{x}, \bar{y})$ so model is linear; fit; inverse-transform back. Why logs work: $\ln$ turns $\times$ into $+$ and powers into $\times$ (e.g. $\ln(Ae^{Bx}) = \ln A + Bx$). Steps: (1) change vars; (2) fit $\bar{y} = a_0 + a_1 \bar{x}$; (3) recover A, B ; (4) check fit.

- $y = Ae^{Bx}$: $\bar{x} = x$, $\bar{y} = \ln y$; $a_0 = \ln A$, $a_1 = B \iff A = e^{a_0}$.
- $y = Ax^n$: $\bar{x} = \ln x$, $\bar{y} = \ln y$; $a_0 = \ln A$, $a_1 = n$.
- $y = A + \frac{B}{x}$: $\bar{x} = \frac{1}{x}$, $\bar{y} = y$; $a_0 = A$, $a_1 = B$.
- $y = \frac{1}{A+Bx}$: $\bar{x} = x$, $\bar{y} = \frac{1}{y}$; $a_0 = A$, $a_1 = B$.
- $y = \frac{1}{B+x}$: $\bar{x} = \frac{1}{x}$, $\bar{y} = \frac{1}{y}$; $a_0 = \frac{1}{A}$, $a_1 = \frac{B}{A} \iff A = \frac{1}{a_0}$, $B = \frac{a_1}{a_0}$.

Trap: not everything linearizes (e.g. $a_0 e^{-x} \sin(a_1 + 2\pi x)$). **Model from graph:** straight $\rightarrow$ linear; explosive growth $\rightarrow Ae^{Bx}$; power curve $\rightarrow Ax^n$; saturating hyperbola $\rightarrow Ax/(B+x)$; reciprocal decay $\rightarrow A+B/x$.

4 3. Grids + Riemann Sums

Partition $a = x_0 < \dots < x_n = b$, $h_i = x_i - x_{i-1}$, sample $x_i^* \in [x_{i-1}, x_i]$: $\int_a^b f \approx \sum f(x_i^*) h_i$. **Uniform:** $h_i \equiv h = \frac{b-a}{n}$, $n+1$ points. **Non-uniform:** varying h_i (adaptive: fine where f changes fast, coarse elsewhere). Idea: spend grid points where the function wiggles; a uniform grid wastes them where f is flat and starves them where f is steep. **Riemann integral** = $\lim_{\max h_j \rightarrow 0} \sum h_i f(x_i^*)$; f bounded, at most countably many jumps (isolated jumps are OK, the area under one point is zero).

5 4. Rectangle / Trapezoid / Simpson

Sample point: $x_i^* = \alpha x_{i-1} + (1-\alpha)x_i$: $\alpha = 1$ left, $\alpha = 0$ right, $\alpha = \frac{1}{2}$ midpoint. (Quiz7: $0.2x_{i-1} + 0.8x_i$ is $\alpha = 0.2$; Lec16: $x_{i-1} + h_i/3$ is $\alpha = \frac{2}{3}$, $x_{i-1} + 2h_i/3$ is $\alpha = \frac{1}{3}$.) General non-uniform: left $\sum h_i f(x_{i-1})$, right $\sum h_i f(x_i)$, mid $\sum h_i f(\frac{x_{i-1}+x_i}{2})$.

$$\bar{I}_{\text{Tr}} = \frac{1}{2} \sum_{i=1}^n \left(f(x_{i-1}) + f(x_i) \right) h_i$$

Uniform trapezoid: $\frac{h(f_0+f_n)}{2} + h \sum_{i=1}^{n-1} f_i$. **Trap idea:** one strip = rectangle hQ_1 + triangle $\frac{h}{2}(Q_2 - Q_1) = h(Q_1 + Q_2)/2$; uniform version averages neighbors (interior f_i counted twice $\rightarrow$ single sum). **Simpson** ($n = 2m$ even, uniform): fits parabola on each pair. Why $O(h^4)$? A parabola matches f 's value, slope AND curvature, so the leftover is two orders smaller than trapezoid (which matches only value + slope). Cost: more f -evaluations per step.

$$I_{\text{parab}} = \frac{h}{3} \left\{ f_0 + f_n + 4 \sum_{\text{odd } i} f_i + 2 \sum_{\text{even } i < n} f_i \right\}$$

Sketch: rectangles = bars at x_i^* ; trapezoids = straight segments node-to-node; Simpson = parabola arcs per pair (blank axes: mark $n+1$ nodes first). Simpson weights: endpoints $\times 1$, odd indices $\times 4$, even interior $\times 2$.

Accuracy estimates

- left/right rectangle $O(h)$; midpoint $O(h^2)$; trapezoid $O(h^2)$; Simpson $O(h^4)$.
- Read $O(h^p)$ as: halving h divides the error by 2^p (rect $\div 2$, mid/trap $\div 4$, Simpson $\div 16$).
- If $|h| < 1$: midpoint beats left/right; trap $\approx$ midpoint order.
- Left-rect global: $|E| \leq \frac{M_1(b-a)}{2} h$, $M_1 = \max |f'|$. Midpoint: $|E| \leq \frac{M_2(b-a)}{24} h^2$. (Taylor + symmetry kills f' term.)
- No elementary antiderivative ($\int e^{x^2} dx$, $\int \frac{dx}{\ln x}$): cross-check Simpson vs trapezoid.

6 5. ODEs: IVP + Numeric Methods

$y' = f(x, y)$: slope field tells the direction at every point; a solution follows the arrows. General solution $y(x, C)$ = whole family of curves; IVP (Cauchy) $+ y(x_0) = y_0$ picks the ONE curve through (x_0, y_0) (fix C by plugging IC; existence/uniqueness assumed). **Check-solution trap:** differentiate candidate AND plug into ODE — both must hold for ALL C . **Bridge (Lec19):** Euler comes from the finite difference $y' \approx \frac{y_{i+1} - y_i}{h}$: replace y' in $y' = f(x, y)$ and solve for y_{i+1} .

$$y_{i+1} = y_i + h f(x_i, y_i), \quad x_{i+1} = x_i + h \quad (\text{Euler})$$

$$y_{i+1} = y_i + h f(x_{i+\frac{1}{2}}, y_{i+\frac{1}{2}}), \quad x_{i+\frac{1}{2}} = x_i + \frac{h}{2}, \quad y_{i+\frac{1}{2}} = y_i + \frac{h}{2} f(x_i, y_i)$$

RK4: weighted average of 4 trial slopes with weights (1, 2, 2, 1)/6: $k_1 = f(x_n, y_n)$; $k_2 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_1)$; $k_3 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_2)$; $k_4 =$

$f(x_n + h, y_n + h k_3)$. $y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$. Price: 4 f -evaluations per step for $O(h^4)$. **Global errors:** Euler $O(h)$ (local $O(h^2) \times N \sim 1/h$); midpoint $O(h^2)$; RK4 $O(h^4)$ (4 evals/step). **Euler error idea:** Taylor $y(x+h) = y(x) + h y'(x) + \frac{h^2}{2} y''(\xi)$; one Euler step from exact $y(x_i)$ misses by $O(h^2)$ (local); $N = (b-x_0)/h = O(1/h)$ steps accumulate to $O(h)$ global. **Midpoint idea:** predict midpoint slope with one Euler half-step, then step full h along it; symmetric $\Rightarrow$ local $O(h^3)$, global $O(h^2)$.

7 6. Linear Algebra* (starred)

det = volume scale factor of the matrix map; det $\neq 0 \iff$ invertible (det = 0 = space collapses). $\det(A \bar{B}) = \det A \det B$. $Ax = b$ has a unique solution $\iff \det A \neq 0 \iff n$ pivots (else 0 or ∞ solutions).

Basic ops: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{pmatrix} = \begin{pmatrix} 17 \\ 39 \end{pmatrix}$ (row $\cdot$ column).

Laplace: expand along row/col with $(-1)^{i+j}$ co-factors (pick the row/col with most zeros). Elimination: row ops change det by sign/scale (swap $\rightarrow -$, scale row by $c \rightarrow \times c$); det = product of U diagonal.

LU: $A = LU$ records elimination (L = unit-lower multipliers, U = upper result); solve $Ly = b$, then $Ux = y$ — two cheap triangular passes instead of re-eliminating. **Worked:** $A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 2 & 4 \end{pmatrix}$: $R_2 - \frac{1}{2} R_1$, $R_3 - \frac{4}{5} R_2 \rightarrow U$ diag (2, 2.5, 3.2), det = $2 \cdot 2.5 \cdot 3.2 = 16$.

8 7. Midterm Recap (compact)

- **Taylor:** $f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n$, $R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}$, $|R_n| \leq \frac{M_{n+1}}{(n+1)!}$.
- Odd fn: only odd powers (deg 8 $\rightarrow x^7$); index R_{2m+1} , bound $\frac{1^{2m+2}}{(2m+2)!}$.
- **Maclaurin:** $e^x, \sin, \cos$ converge $\forall \mathbb{R}$; $\frac{1}{1-x} = \sum x^k, \ln(1+x), \arctan$ for $|x| < 1$.
- **Horner:** tableau at c gives $p(c)$ (last) + quotient q ; tableau on q gives $p'(c)$.
- **Interpolation:** $\ell_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$, $p = \sum y_i \ell_i$; err = $\frac{f^{(n+1)}(\xi)}{(n+1)!} \prod (x - x_i)$ ($= 0$ at nodes).
- Vandermonde det = $\prod_{i < j} (x_j - x_i) \neq 0$ (distinct nodes); DD: $f[x_i, x_{i+1}] = \frac{\Delta f}{\Delta x}$, Newton $p_n = f[x_0] + \sum f[x_0..x_k] \prod_{j < k} (x - x_j)$.

9 8. Worked Mini-Examples

Coffee regression (Quiz 6)

Given $x: 0, 1, 2, 3$; $y: 3, 5, 8, 7$ ($n+1=4$ pts). Step 1 — sums (add term by term):

$$\begin{aligned} \sum x_i &= 0+1+2+3 = 6; \quad \sum y_i = 3+5+8+7 = 23; \\ \sum x_i^2 &= 0+1+4+9 = 14; \\ \sum x_i y_i &= 0+5+16+21 = 42. \end{aligned}$$

Step 2 — means (divide by 4): $\bar{x} = 6/4 = 1.5$; $\bar{y} = 23/4 = 5.75$; $\bar{x}^2 = 14/4 = 3.5$; $\overline{xy} = 42/4 = 10.5$. Step 3 — coeffs (plug into boxed formula): $a_1 = \frac{10.5 - 1.5 \cdot 5.75}{3.5 - 1.5^2} = \frac{10.5 - 8.625}{3.5 - 2.25} = \frac{1.875}{1.25} = 1.5$; $a_0 = 5.75 - 1.5 \cdot 1.5 = 5.75 - 2.25 = 3.5$. $\hat{y} = 3.5 + 1.5x$. Step 4 —

predict $x=4$: $\hat{y} = 3.5 + 1.5 \cdot 4 = 3.5 + 6 = 9.5$ (mark (4,9.5) on the scatter plot with the line).

Plot recipe (Quiz6 Q2): scatter the 4 data points, draw the line through (0,3.5) and (4,9.5) — regression line need NOT pass through data points. Step 5 — check fit: $\hat{y}: 3.5, 5.0, 6.5, 8.0$ vs $y: 3, 5, 8, 7$; residuals $-0.5, 0, 1.5, -1.0$; squared error $S = 0.25 + 0 + 2.25 + 1 = 3.5$ (this S is what least squares minimized). **Battery prediction:** at $t=70$: $0.714 \cdot 70 + 3.57 = 53.55\%$ (extrapolation — flag as less reliable). **Hessian:** from Step 1 sums ($\sum x_i = 6$, $\sum x_i^2 = 14$): $S_{a_0 a_0} = 2 \cdot 4 = 8$, $S_{a_0 a_1} = 2 \cdot 6 = 12$, $S_{a_1 a_1} = 2 \cdot 14 = 28$; $\det H = 8 \cdot 28 - 12^2 = 224 - 144 = 80 > 0$ (+ $8 > 0$) $\Rightarrow$ min.

Battery fit (Lec 14)

Given $t: 0, 10, 20, 30, 40, 50, 60$; charge 5, 10, 15, 25, 35, 40, 45 ($n+1=7$). Sums: $\sum t_i = 0+10+20+30+40+50+60 = 210$; $\sum t_i^2 = 5+10+15+25+35+40+45 = 175$; $\sum t_i^3 = 0+100+400+900+1600+2500+3600 = 9100$; $\sum t_i y_i = 0+100+300+750+1400+2000+2700 = 7250$. Means (divide by 7): $\bar{x} = 210/7 = 30$;

$$\bar{y} = 175/7 = 25; \bar{x^2} = 9100/7 = 1300;$$

$$\overline{xy} = 7250/7 \approx 1035.71. \text{ Coeffs:}$$

$$a_1 = \frac{1035.71 - 30 \cdot 25}{1300 - 30^2} = \frac{1035.71 - 750}{1300 - 900} = \frac{285.71}{400} \approx 0.714; a_0 = 25 - 0.714285 \cdot 30$$

$$= 25 - 21.43 \approx 3.57. \quad \boxed{\phi(x) = 0.714x + 3.57}.$$

Check fit:

$$\hat{y} : 3.57, 10.71, 17.85, 24.99, 32.13, 39.27, 46.41 \text{ vs}$$

$$y : 5, 10, 15, 25, 35, 40, 45$$

(residuals)

$$1.43, -0.71, -2.85, 0.01, 2.87, 0.73, -1.41).$$

Exp linearization (HW09)

Given $x : 1, 2, 3, 5, y : 8, 4, 2, 1$; model $y = Ae^{Bx} \Rightarrow$ transform $\tilde{x} = x, \tilde{y} = \ln y$. Step 0 — transform each y : $\ln 8 \approx 2.07944, \ln 4 \approx 1.38629, \ln 2 \approx 0.69315, \ln 1 = 0$. So $\tilde{x} : 1, 2, 3, 5; \tilde{y} : 2.07944, 1.38629, 0.69315, 0$. Step 1 — means ($n+1=4$): $\bar{\tilde{x}} = (1+2+3+5)/4 = 11/4 = 2.75; \bar{\tilde{y}} = (2.07944+1.38629+0.69315+0)/4 = 4.15888/4 \approx 1.03972$ (note $= \ln 64/4 = 1.5 \ln 2$); $\bar{\tilde{x}^2} = (1+4+9+25)/4 = 39/4 = 9.75; \bar{\tilde{x}\tilde{y}} = (1 \cdot 2.07944 + 2 \cdot 1.38629 + 3 \cdot 0.69315 + 5 \cdot 0)/4 = 6.93147/4 \approx 1.73287$ (note $= 2.5 \ln 2$). Step 2 — fit: $a_1 = \frac{1.73287 - 2.75 \cdot 1.03972}{9.75 - 2.75^2} = \frac{1.73287 - 2.85923}{9.75 - 7.5625} = \frac{-1.12636}{2.1875} \approx -0.5149;$

$$a_0 = 1.03972 - (-0.5149) \cdot 2.75$$

$$= 1.03972 + 1.41598 \approx 2.4557. \text{ Step 3 —}$$

back-transform ($A = e^{a_0}, B = a_1$):

$$\hat{y} = e^{2.4557} \approx 11.65, B \approx -0.515 < 0. \text{ Check:}$$

$$\hat{y} = 11.65e^{-0.515x} \text{ gives } 7.0, 4.2, 2.5, 0.9 \text{ vs } 8, 4, 2, 1$$

— reasonable. **Nonlinear example (Lec15):**

$$x : 1, 2, 3, 4, y : 0.45, 0.81, 1.05, 1.13 \text{ for } y = \frac{Ax}{B+x}:$$

$$\tilde{x} : 1, 2, 3, \tilde{y} = 1/y \text{ give } a_1 \approx 1.834, a_0 \approx 0.368,$$

$$\text{so } A = 1/a_0 \approx 2.71, B = a_1/a_0 \approx 4.98.$$

Trapezoid $\sin x$ (Quiz 7)

Given $[0, 3\pi/2]$, uniform $h = \pi/4 \Rightarrow$

$n = (3\pi/2)/(\pi/4) = 6$ subintervals, 7 nodes

$$x_i = i\pi/4. \text{ Step 1 — evaluate } f(x) = \sin x:$$

$$\sin 0 = 0; \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} \approx 0.70711; \sin \frac{\pi}{2} = 1;$$

$$\sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2}; \sin \pi = 0; \sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2};$$

$$\sin \frac{3\pi}{2} = -1. \text{ Step 2 — uniform trapezoid}$$

(endpoints half-weighted):

$$\tilde{I} = \frac{h}{2}(f_0 + f_6) + h \sum_{i=1}^5 f_i \text{ with}$$

$$\sum_{i=1}^5 f_i = \frac{\sqrt{2}}{2} + 1 + \frac{\sqrt{2}}{2} + 0 - \frac{\sqrt{2}}{2} = 1 + \frac{\sqrt{2}}{2} \approx 1.70711.$$

$$\text{Step 3 — multiply out: } \tilde{I} = \frac{h}{2}(f_0 + f_6) + h \sum_{i=1}^5 f_i$$

$$= \frac{\pi}{4} \left(\frac{1}{2}(-1) + 1.70711 \right) = \frac{\pi}{4}(1.20711) \approx$$

$$0.78540 \cdot 1.20711 \approx 0.94806. \text{ Step 4 — exact}$$

$$I = \int_0^{3\pi/2} \sin x \, dx = [-\cos x]_0^{3\pi/2} =$$

$$-\cos \frac{3\pi}{2} + \cos 0 = 0 + 1 = 1; \text{ err}$$

$$= |1 - 0.94806| \approx 0.05194. \text{ How to draw}$$

trapezoids (Pract.6): (1) mark all $n+1$ nodes on the axis;

(2) plot $(x_i, f(x_i))$; (3) join consecutive points with straight segments; (4) each strip is a trapezoid (drop verticals to the axis). Parabolae (Simpson): same nodes, one arc per pair through its 3 points.

Simpson $\int_2^4 \frac{dx}{\ln x}$ (Quiz 8)

Given $n = 8$ (even $\checkmark$), 9 points,

$$h = (4 - 2)/8 = 0.25. \text{ Nodes } x_i = 2 + 0.25i:$$

$$2, 2.25, 2.5, 2.75, 3, 3.25, 3.5, 3.75, 4. \text{ Step 1 —}$$

evaluate $f(x) = 1/\ln x$ at each node:

$$f_0 = 1/\ln 2 \approx 1.44270; f_1 = 1/\ln 2.25 \approx 1.23315;$$

$$f_2 = 1/\ln 2.5 \approx 1.09136; f_3 = 1/\ln 2.75 \approx 0.98853;$$

$$f_4 = 1/\ln 3 \approx 0.91024; f_5 = 1/\ln 3.25 \approx 0.84842;$$

$$f_6 = 1/\ln 3.5 \approx 0.79824; f_7 = 1/\ln 3.75 \approx 0.75657;$$

$$f_8 = 1/\ln 4 \approx 0.72135. \text{ Step 2 — weight (ends } \times 1,$$

$$\text{odd } i \times 4, \text{ even } i \times 2): 4 \sum_{\text{odd}} =$$

$$4(1.23315 + 0.98853 + 0.84842 + 0.75657) =$$

$$4(3.82667) = 15.30668;$$

$$2 \sum_{\text{even}} = 2(1.09136 + 0.91024 + 0.79824) =$$

$$2(2.79984) = 5.59968. \text{ Step 3 — assemble:}$$

$$I = \frac{0.25}{3}(1.44270 + 15.30668 + 5.59968 + 0.72135)$$

$$= \frac{0.25}{3}(23.07041) \approx 0.083333 \cdot 23.07041 \approx 1.92253.$$

Euler + midpoint $y' = x - 4y, y(0) = -3/4$ (Quiz 9)

$$h = 0.2, f(x, y) = x - 4y. \text{ Euler:}$$

$$f_0 = 0 - 4(-0.75) = 3;$$

$$y_1 = -0.75 + 0.2(3) = -0.15;$$

$$f_1 = 0.2 - 4(-0.15) = 0.8:$$

$$y_2 = -0.15 + 0.2(0.8) = 0.01;$$

$$f_2 = 0.4 - 0.04 = 0.36;$$

$$y_3 = 0.01 + 0.2(0.36) = 0.082. \text{ Midpoint:}$$

$$\text{half-step } (0.1, -0.75 + 0.1(3)) = (0.1, -0.45),$$

$$f = 0.1 + 1.8 = 1.9$$

$$\rightarrow y_1 = -0.75 + 0.2(1.9) = -0.37; \text{ half-step}$$

$$(0.3, -0.37 + 0.1(1.68)) = (0.3, -0.202),$$

$$f = 0.3 + 0.808 = 1.108$$

$$\rightarrow y_2 = -0.37 + 0.2(1.108) = -0.1484. \text{ Exact}$$

$$y(0.4) \approx -0.101304 \text{ (given in quiz; from closed}$$

$$\text{form } y = x/4 - 1/16 - \frac{11}{16}e^{-4x}): \text{ Euler err}$$

$$= |-0.101304 - 0.01| \approx 0.11130; \text{ midpoint err}$$

$$= |-0.101304 - (-0.1484)| \approx 0.04710. \text{ IVP from}$$

$$\text{general sol: } y = -\frac{1}{4} + \frac{x}{2} + Ce^{-2x},$$

$$(x_0, y_0) = (\frac{1}{2}, \frac{1}{4}): \frac{1}{4} = -\frac{1}{4} + \frac{1}{4} + Ce^{-1} \Rightarrow$$

$$C = e/4 \approx 0.6796, y^* = -\frac{1}{4} + \frac{x}{2} + \frac{e}{4}e^{-2x}. \text{ HW10}$$

$$\text{Q2 pattern: } y' = \frac{x}{2} + 2x^2, (1, 4), \text{ general}$$

$$y = (x^2 + C)x: 4 = (1 + C) \cdot 1 \Rightarrow C = 3,$$

$$y^* = (x^2 + 3)x. \text{ Euler pattern: } x_1 = 1 + h,$$

$$y_1 = 4 + h(4/1 + 2) = 4 + 6h \text{ (since } f(1, 4) = 6).$$

$$\text{Midpoint pattern: } x_{1/2} = 1 + h/2,$$

$$y_{1/2} = 4 + (h/2) \cdot 6 = 4 + 3h. \text{ Check trap:}$$

$$y = Ce^x - 2 \cos 2x - 2 \sin 2x \text{ is not a sol of}$$

$$y' = y + 5 \sin 2x: y' = Ce^x + 4 \sin 2x - 4 \cos 2x,$$

$$\text{but } y + 5 \sin 2x = Ce^x - 2 \cos 2x + 3 \sin 2x;$$

$$\text{difference} = \sin 2x - 2 \cos 2x \neq 0.$$

$$\text{But } y = \frac{1}{2}(\cos x - \sin x) + Ce^x \text{ is a sol of}$$

$$y' = y - \cos x:$$

$$y' = \frac{1}{2}(-\sin x - \cos x) + Ce^x = y - \cos x.$$

Rule: differentiate candidate AND plug into ODE

— both must hold $\forall C$. **Decision boundary**

mini-example: plane $-1 + x_1 + x_2 = 0$: point

$(0, 0)$ gives $-1 < 0$ (class 1), $(1, 1)$ gives $+1 > 0$

(class 2); $\sigma(-1 + x_1 + x_2)$ maps these to ≈ 0.27 and

≈ 0.73 . **RK4 schemes (no numbers needed):**

$$\text{for } y' = y + x: k_1 = y_i + x_i, k_2 = y_i + \frac{h}{2}k_1 + x_i + \frac{h}{2},$$

$$k_3 = y_i + \frac{h}{2}k_2 + x_i + \frac{h}{2}, k_4 = y_i + hk_3 + x_i + h.$$

$$\text{For } y' = x(y + 1/x) \text{ (} h=1/12\text{): } k_1 = x_i y_i + 1/x_i;$$

$$k_2 = (x_i + \frac{1}{24})(y_i + \frac{k_1}{24}) + 1/(x_i + \frac{1}{24}); k_3 \text{ same with}$$

$$k_2; k_4 = (x_i + \frac{1}{12})(y_i + \frac{k_3}{12}) + 1/(x_i + \frac{1}{12}).$$

Rectangle lattice nodes (Quiz 7)

$$\text{Given } x_k = \frac{3 \ln(1+k/5)}{\ln 2} - 1, k = 0..5. \text{ Sample:}$$

$$x_1 = \frac{3 \ln 1.2}{\ln 2} - 1 \approx -0.21090 \text{ (since}$$

$$3 \ln 1.2 \approx 0.54696, \ln 2 \approx 0.69315);$$

$$x_0 = \frac{3 \ln 1}{\ln 2} - 1 = -1; x_5 = \frac{3 \ln 2}{\ln 2} - 1 = 2.$$

$$\text{Full list: } -1.00000, -0.21090, 0.45628, 1.03422,$$

$$1.54399, 2.00000. x_i^* = 0.2x_{i-1} + 0.8x_i: -0.36872,$$

$$0.32285, 0.91863, 1.44204, 1.90880.$$

$$\text{Rect } i \text{ has width } h_i = x_i - x_{i-1}, \text{ height } f(x_i^*)$$

$$(f = x^3 - 1.25x^2). \text{ How to draw (Pract.5): (1)}$$

mark the $n+1$ nodes on the x -axis; (2) for each

subinterval compute x_i^* using the α -formula (any α :

$1/3, 2/3, 0.2$, midpoint); (3) go up to the curve at

$f(x_i^*)$ — that height is the bar top; (4) draw the

bar over $[x_{i-1}, x_i]$ (works below the axis too:

height negative $\Rightarrow$ signed area).

Accuracy table $\int_0^1 x^2 dx$ (Lec17, $h = 0.25$)

$$\text{Nodes } 0, \frac{1}{4}, \frac{3}{4}, 1; f(x) = x^2 \text{ gives } 0, \frac{1}{16}, \frac{9}{16}, 1.$$

$$\text{Left} = \frac{17}{64} \text{ ? No: } f_3 = (\frac{3}{4})^2 = \frac{9}{16}, \text{ not } \frac{3}{4}. \text{ Correct}$$

$$\text{left uses } f_0..f_3: = \frac{1}{4}(0 + \frac{1}{16} + \frac{1}{4} + \frac{9}{16}) = \frac{1}{4}(\frac{14}{16})$$

$$= \frac{7}{32} \approx 0.21875, \text{ err}$$

$$= |0.33333 - 0.21875| \approx 0.11458.$$

$$\text{Mid: midpoints } \frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}, f^* : \frac{1}{64}, \frac{9}{64}, \frac{25}{64}, \frac{49}{64};$$

$$\tilde{I} = \frac{1}{4} \cdot \frac{84}{64} \approx 0.32813 \text{ (err } 0.00521).$$

$$\text{Right} = \frac{1}{4}(\frac{1}{16} + \frac{1}{4} + \frac{9}{16} + 1) = \frac{15}{32} \approx 0.46875 \text{ (err}$$

$$0.13542).$$

$$\text{Trap} = \frac{11}{32} \approx 0.34375 \text{ (err } 0.01042). \text{ Derivation}$$

idea (Th.10): Taylor

$$f(x) = f(x_{i-1}) + f'(\xi_x)(x - x_{i-1}); \text{ integrate over}$$

$$[x_{i-1}, x_i]: \text{ local err } |E_i| \leq M_1 h_i^2/2; \text{ sum:}$$

$$|E| \leq \frac{M_2}{2} \sum h_i^2; \text{ uniform: } \sum h_i^2 = nh^2,$$

$$nh = b - a$$

$$\Rightarrow |E| \leq \frac{M_2(b-a)}{2} h = O(h). \text{ Midpoint: expand at}$$

$$c_i; \text{ odd term } \int (x - c_i) = 0 \text{ vanishes } \Rightarrow O(h^2).$$

Simpson $\int_1^2 e^{x^2} dx$ (Lec19, $n = 4$)

$$\text{Given } h = (2 - 1)/4 = 0.25, \text{ nodes } x_i = 1 + 0.25i:$$

$$1, \frac{5}{4}, \frac{3}{2}, \frac{7}{4}, 2. \text{ Step 1 — evaluate } f(x) = e^{x^2} \text{ (square}$$

$$x \text{ first, then exponentiate): } f_0 = e^1 \approx 2.71828;$$

$$f_1 = e^{(5/4)^2} = e^{1.5625} \approx 4.77073;$$

$$f_2 = e^{(3/2)^2} = e^{2.25} \approx 9.48774;$$

$$f_3 = e^{(7/4)^2} = e^{3.0625} \approx 21.38094;$$

$$f_4 = e^4 \approx 54.59815.$$

$$I = \frac{0.25}{3}(2.71828 + 4(26.15167) + 2(9.48774) +$$

$$54.59815) \approx 15.0749.$$

Trap check: two double-intervals $T_1 = h(f_0 + f_2) =$

$$0.25(2.71828 + 9.48774) = 0.25(12.20602) \approx 3.0515,$$

$$T_2 = h(f_2 + f_4) = 0.25(9.48774 + 54.59815) =$$

$$0.25(64.08589) \approx 16.0215, \text{ total } \approx 19.0730 \rightarrow 19.07$$

— same ballpark $\Rightarrow$ likely right. **Simpson sketch**

(Quiz8 Q1): $[0, 240]$, 6 subintervals $\Rightarrow h = 40$,

nodes $0, 40, \dots, 240$; one parabola arc per pair of

subintervals (3 arcs total); mark nodes, draw each

arc through its 3 points.

Pseudoinverse $\rightarrow$ 1D formulas (HW09)

$$X^T X \bar{a} = X^T \bar{y} \text{ with } X = [1 \ x_i] \text{ gives}$$

$$\left(\sum_{i=1}^{n+1} x_i \sum_{i=1}^{n+1} x_i^2 \right) \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{$$